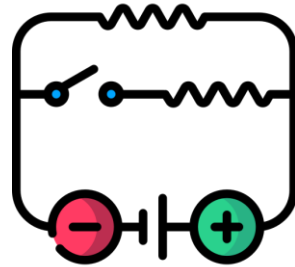


EMJ12103

ELECTRIC CIRCUIT 1

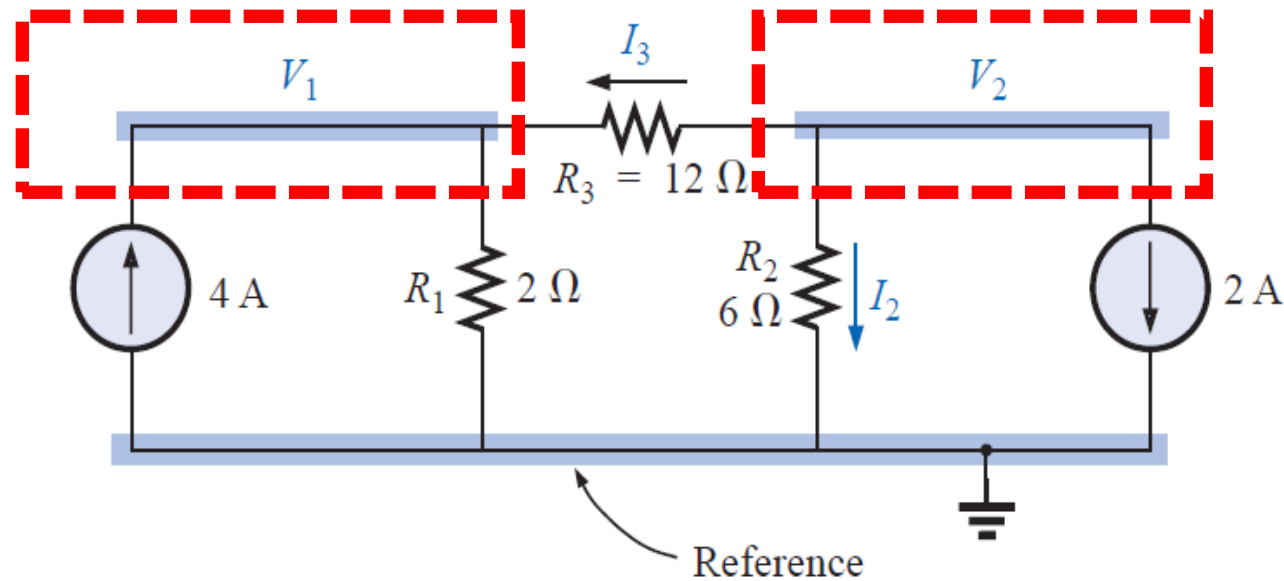


Chapter 3: Nodal Analysis

DR. AZRALMUKMIN AZMI

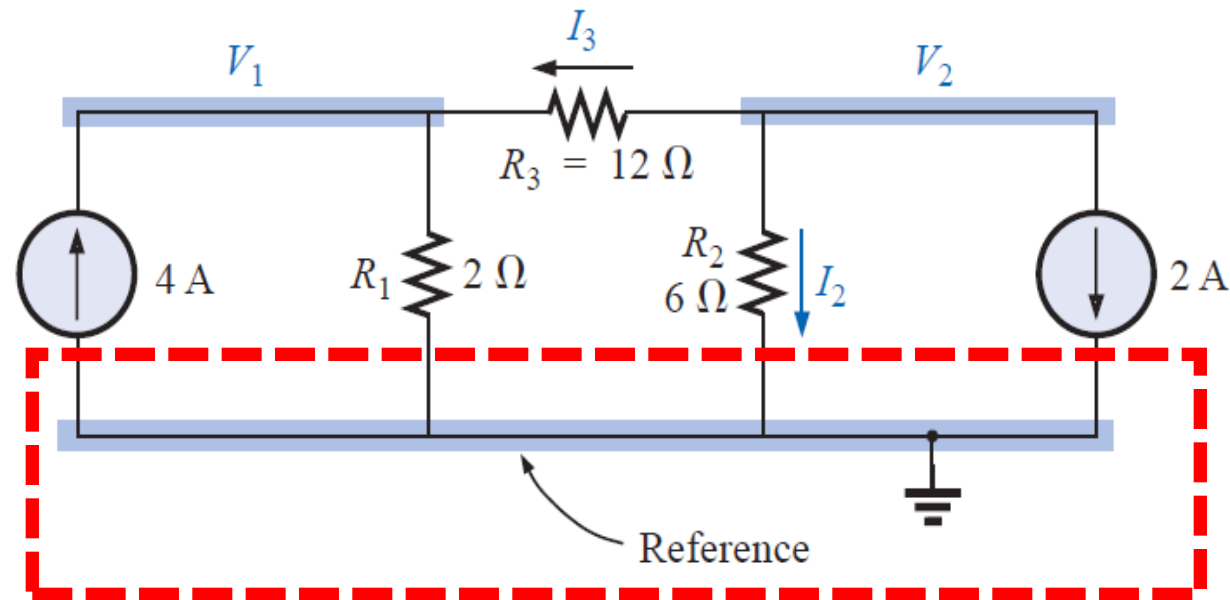
Basic Concepts

- ▶ All the points in a circuit joined together with **zero resistance** wires make up a **single node**. The **voltage** is the **same** at any point in a node.



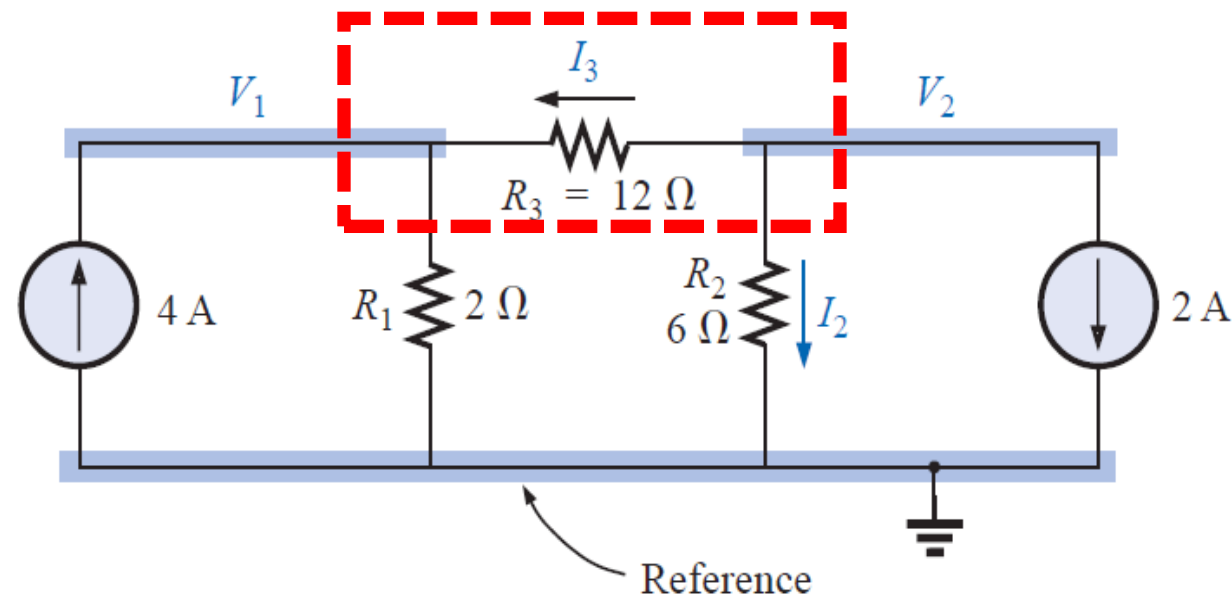
Basic Concepts (Cont...)

- Choose one node in a circuit to be **reference node**. Measure all node voltages relative to this one.



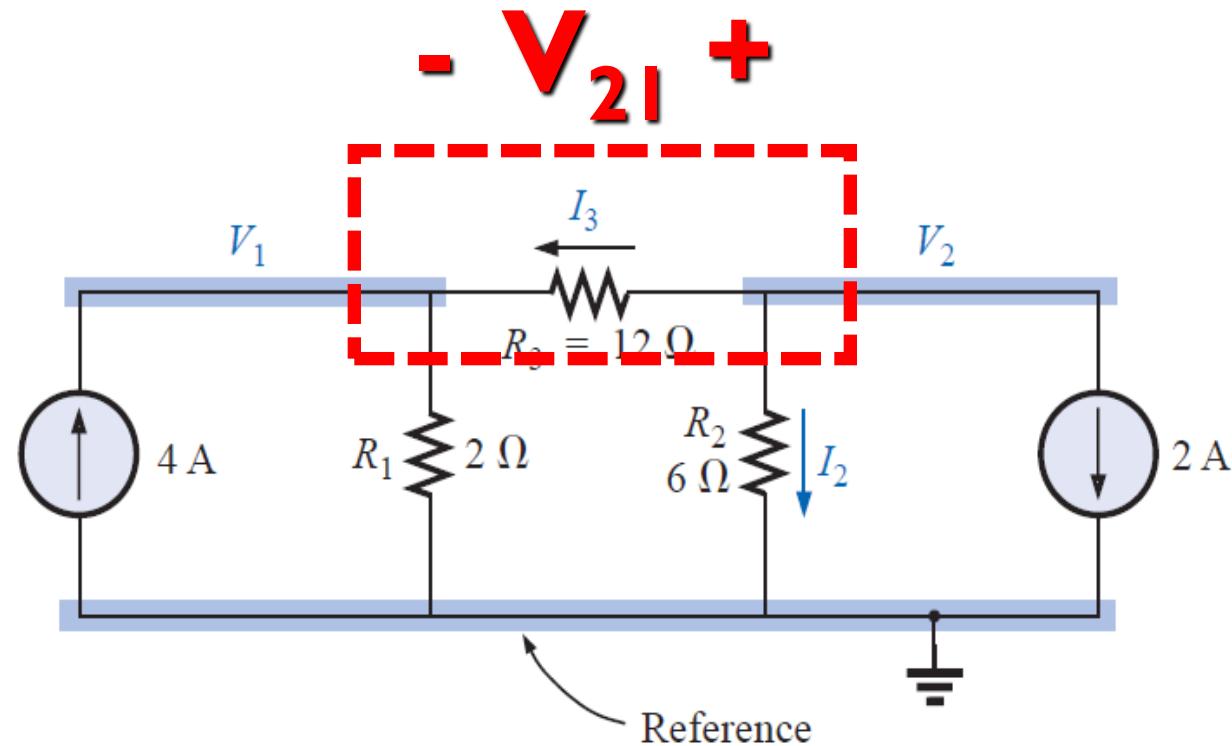
Basic Concepts (Cont...)

- ▶ Voltage across a component is the difference between the voltages at the two ends. (e.g $V_{21} = V_2 - V_1$)



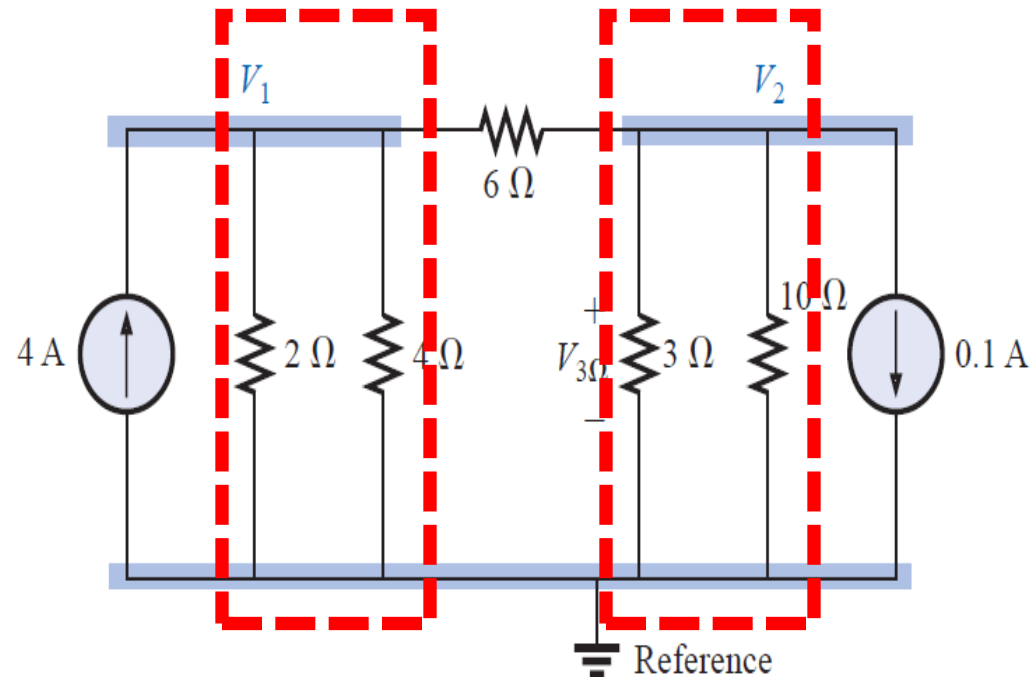
Basic Concepts (Cont...)

- ▶ Passive sign convention: **Current** and **voltage** arrows for a 2 terminal component are in **opposite directions**.



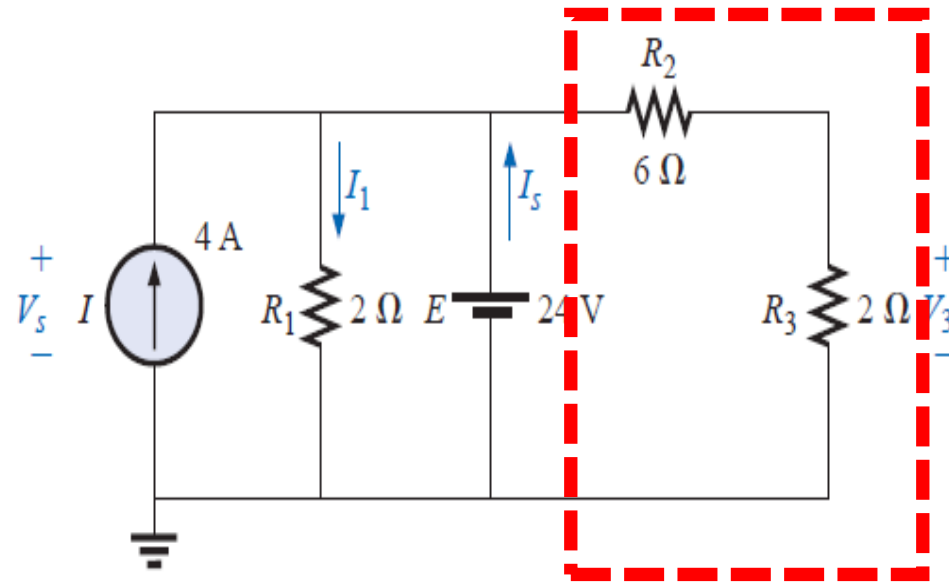
Basic Concepts (Cont...)

- ▶ Components are **in parallel** if they are joined together at both ends.
- ▶ Components in parallel have the **same voltage** across them.



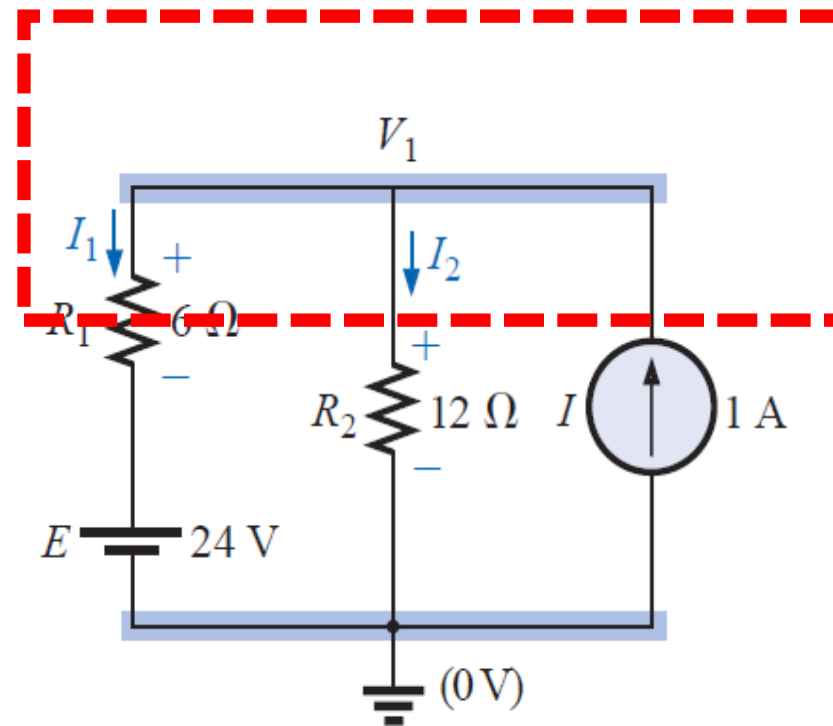
Basic Concepts (Cont...)

- Components are **in series** if the **same current** is forced to flow through each one.



Basic Concepts (Cont...)

- KCL (Kirchoff's Current Law): The **sum of currents entering** a node is **equal** to the **sum of the currents leaving** a node.



Nodal Analysis

STEP 1:

Select a node as reference node and assign V_1 and V_2 and so on to the remaining nodes

STEP 2:

Apply KCL at each node in the circuit. Use Ohm's Law to express the branch currents in term of node voltage.

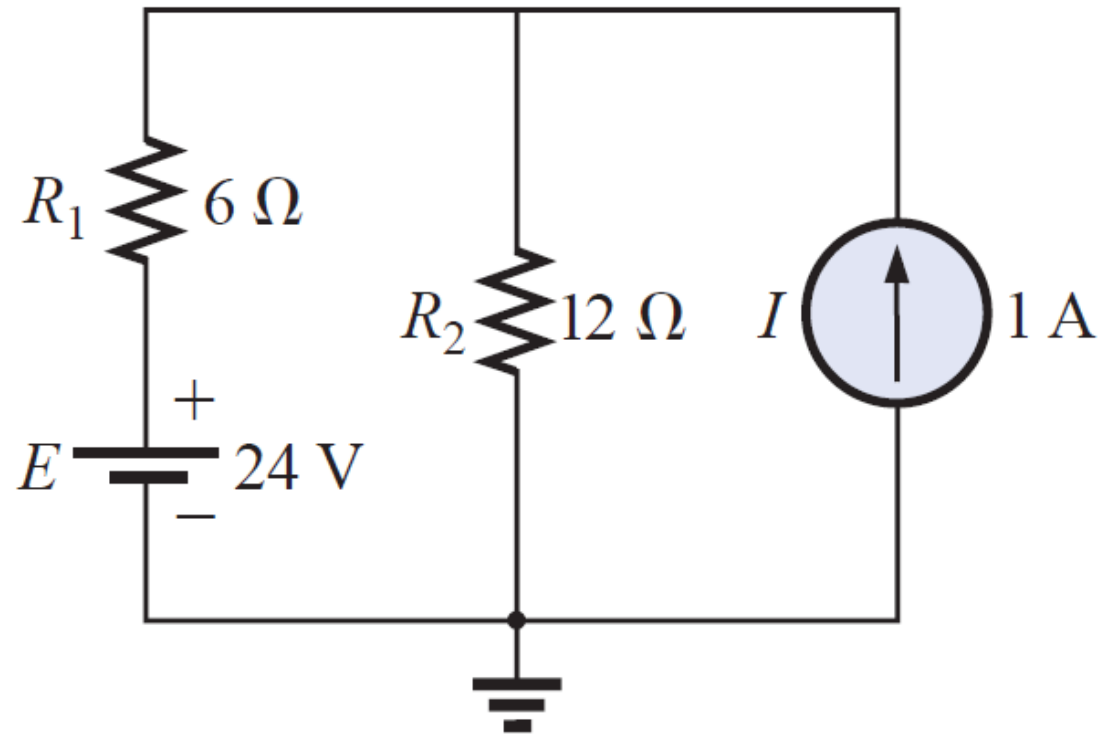
STEP 3:

Solve the resulting simultaneous equation to obtain unknown node voltage



Nodal Analysis (Cont...)

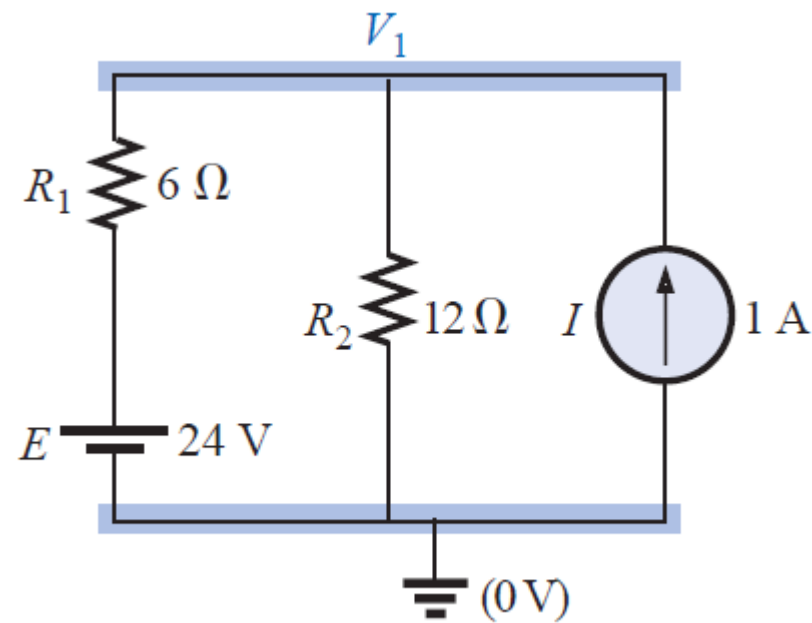
- Apply nodal analysis to the network:



Nodal Analysis (Cont...)

STEP 1:

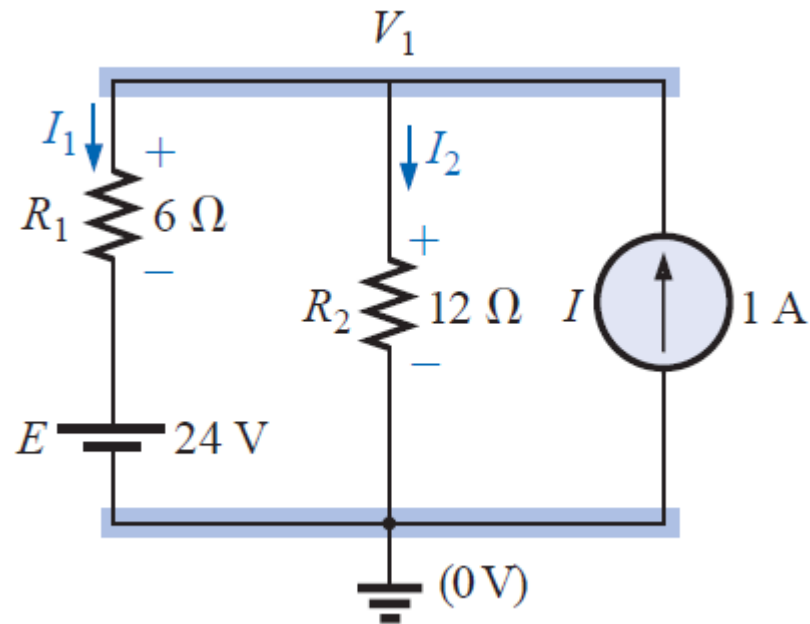
Select a node as reference node and assign V_1 and V_2 and so on to the remaining nodes



Nodal Analysis (Cont...)

STEP 2:

Apply KCL at each node in the circuit. Use Ohm's Law to express the branch currents in term of node voltage.



Using KCL at node V_1

$$I = I_1 + I_2$$

$$I_2 = \frac{V_1 - 0}{R_2} = \frac{V_1}{12}$$

$$I_1 = \frac{V_1 - E}{R_1} = \frac{V_1 - 24}{6}$$

Nodal Analysis (Cont...)

STEP 3:

Solve the resulting simultaneous equation to obtain unknown node voltage

Substitute into KCL at node V_1 :

$$I = I_1 + I_2$$

Then :

$$1 = \frac{V_1 - 24}{6} + \frac{V_1}{12}$$

$$I_1 = \frac{V_1 - 24}{6} = \frac{20 - 24}{6} = -0.6667A$$

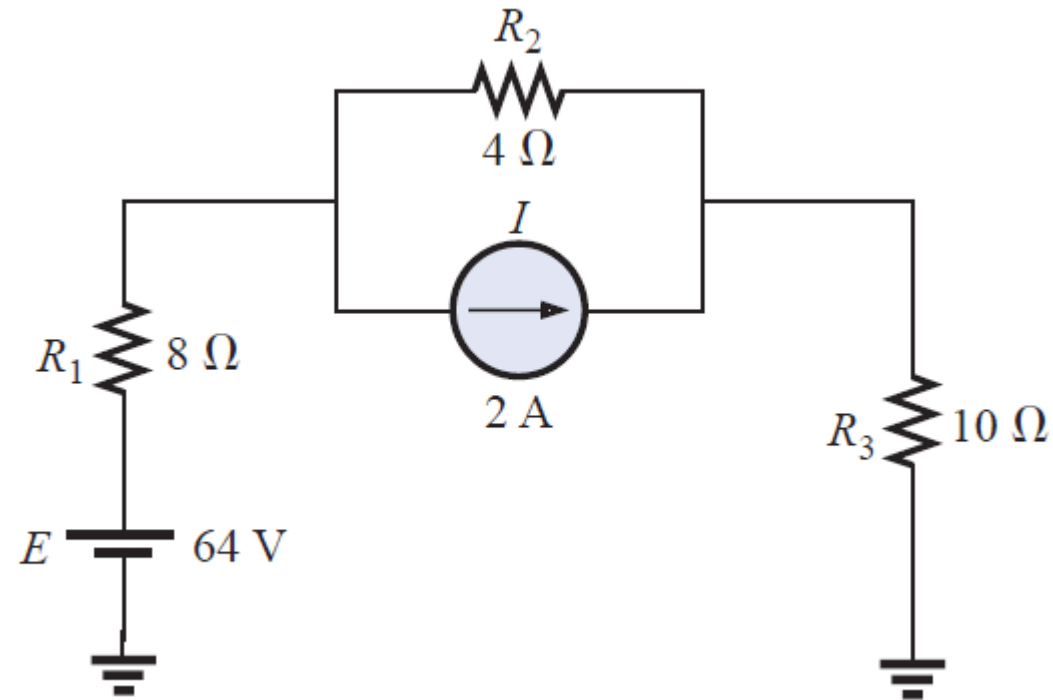
$$V_1 = 20V$$

$$I_2 = \frac{V_1}{12} = \frac{20}{12} = 1.6667A$$



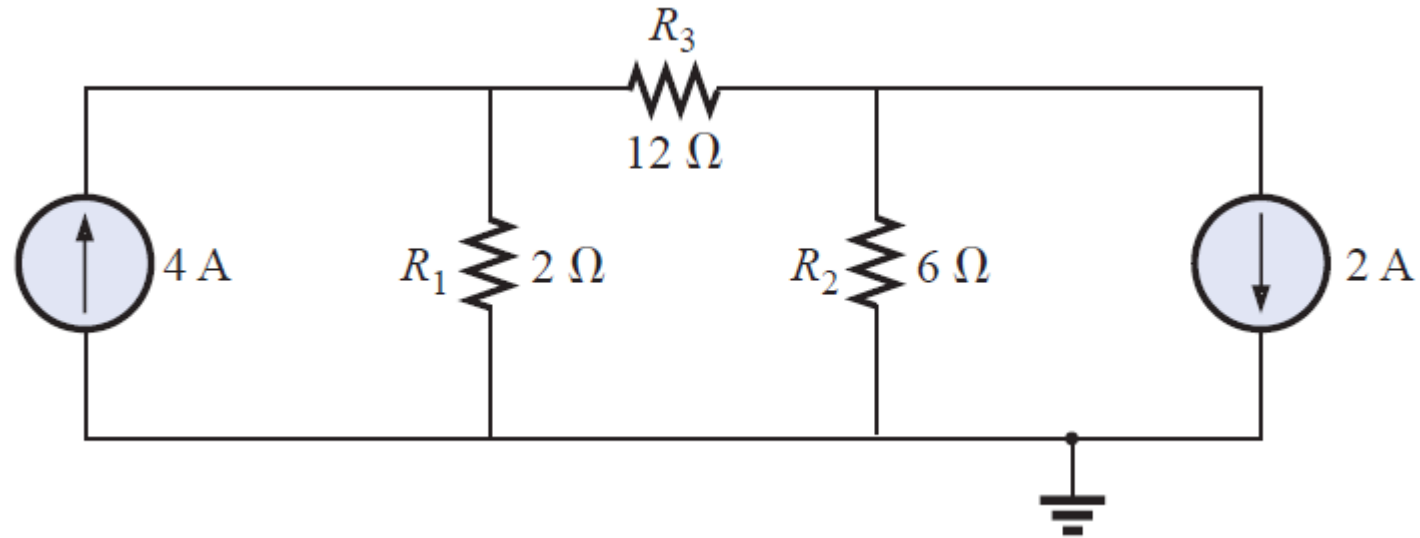
Nodal Analysis (Cont...)

- Apply nodal analysis to the network:



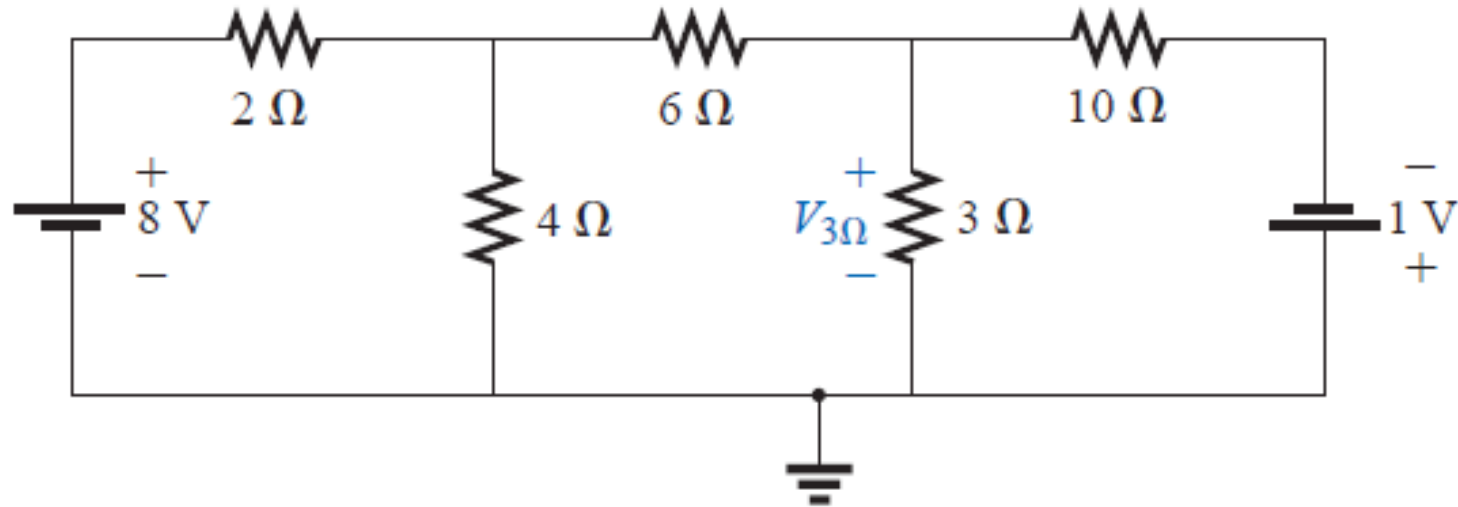
Nodal Analysis (Cont...)

- Determine the nodal voltages for the network:



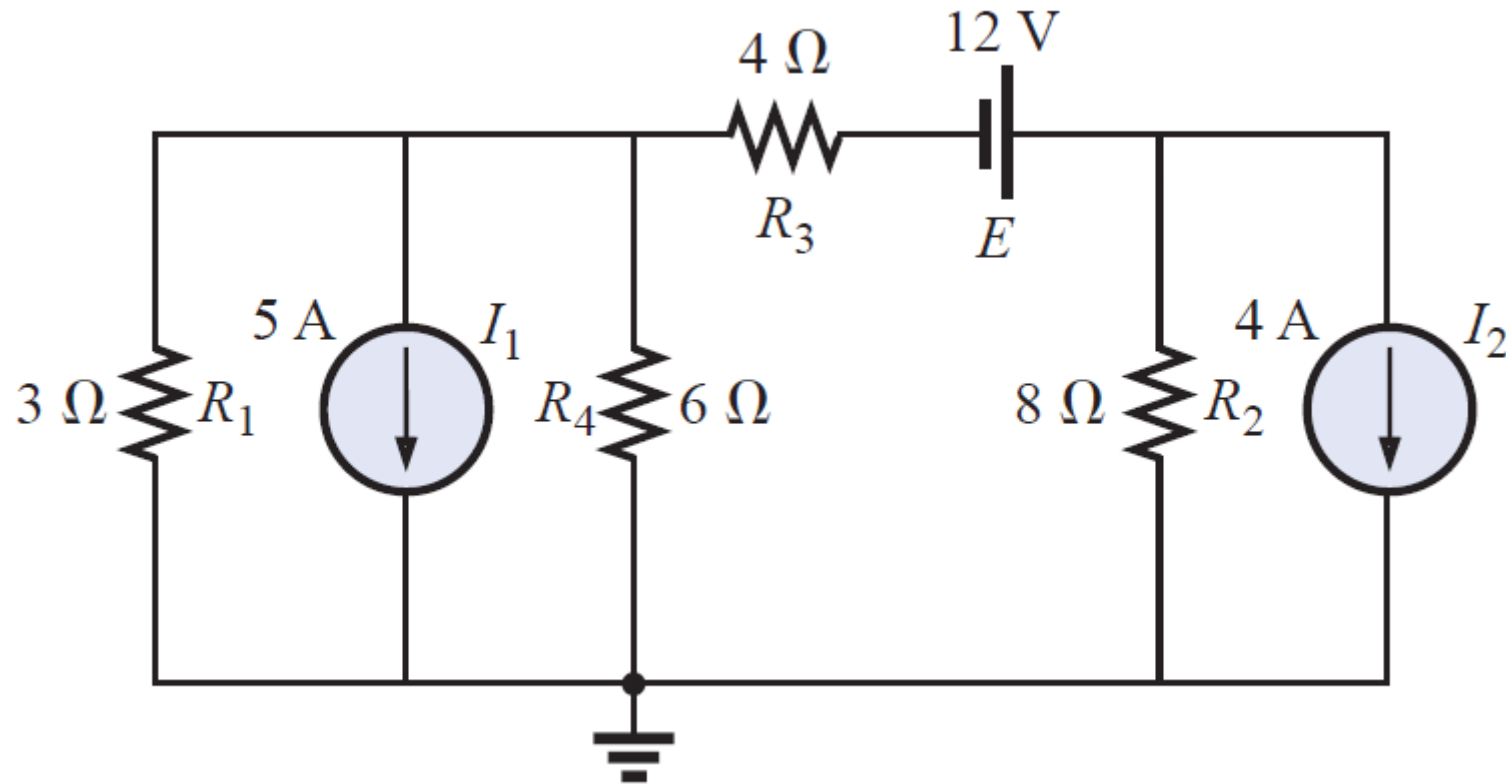
Nodal Analysis (Cont...)

- Find voltage across the $3\ \Omega$:



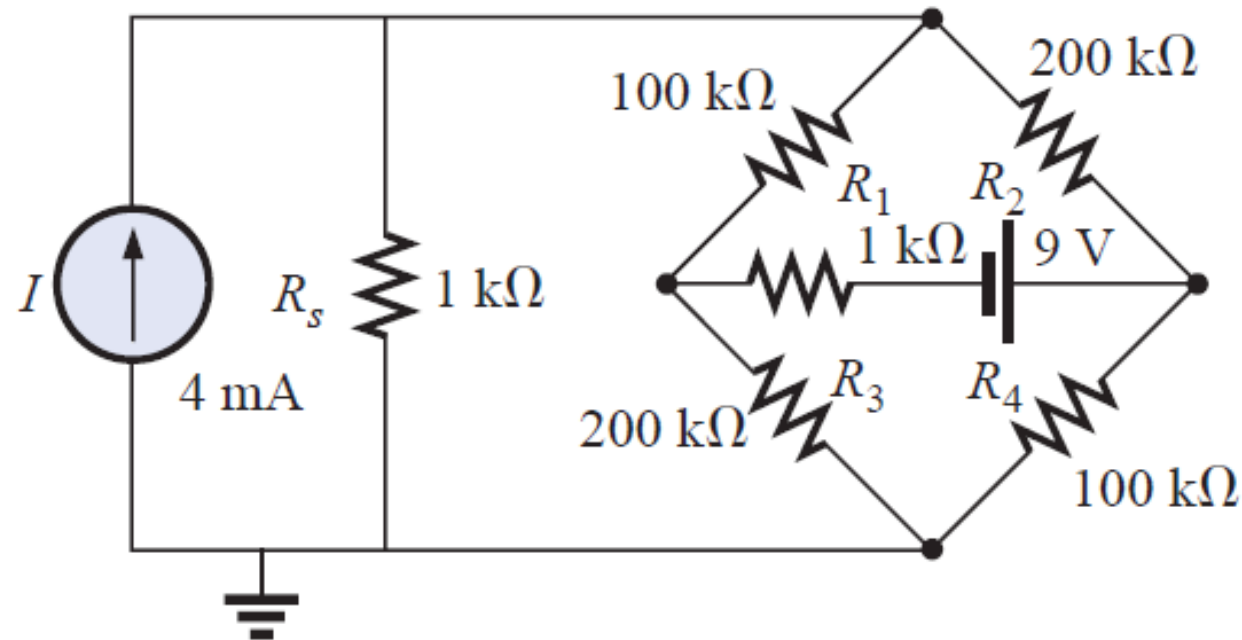
Nodal Analysis (Cont...)

- Determine the nodal voltages for the network:



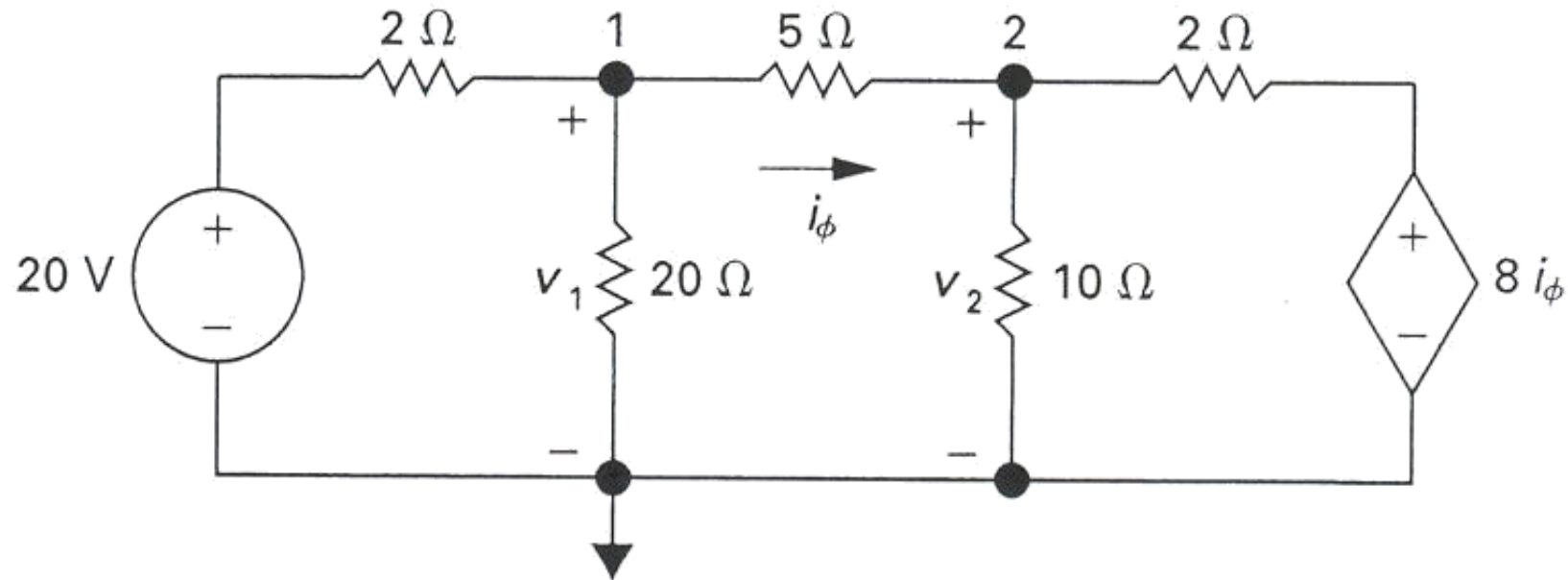
Nodal Analysis (Cont...)

- Find current across the R_S :



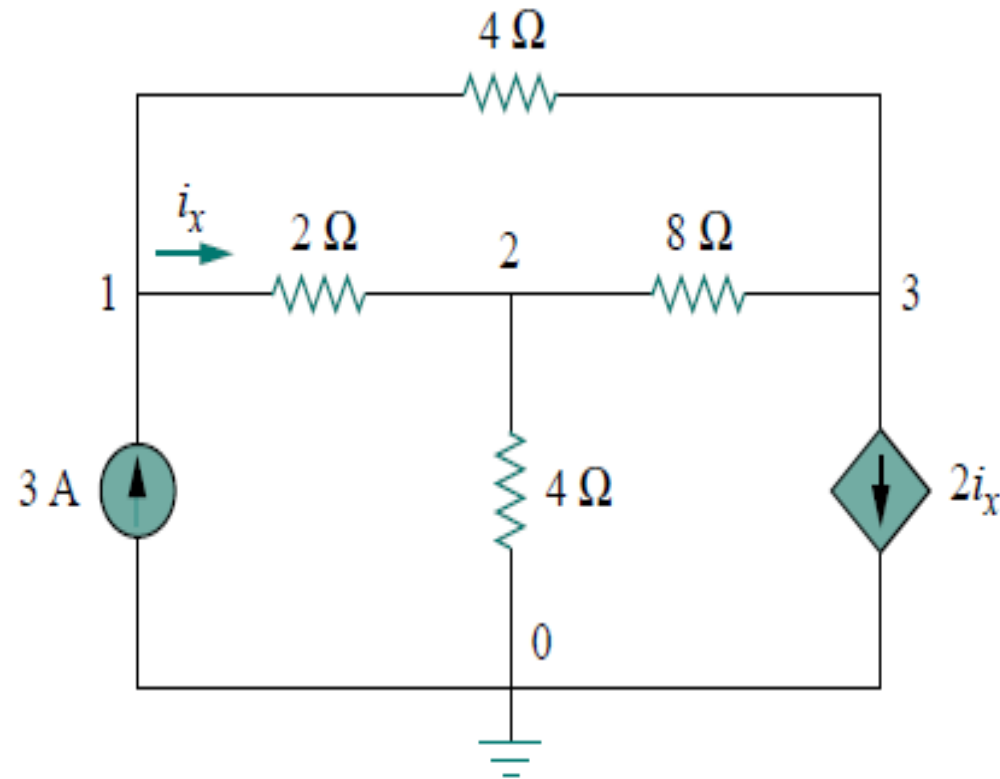
Nodal Analysis (Cont...)

- Calculate the node voltages in the circuit shown in figure below:

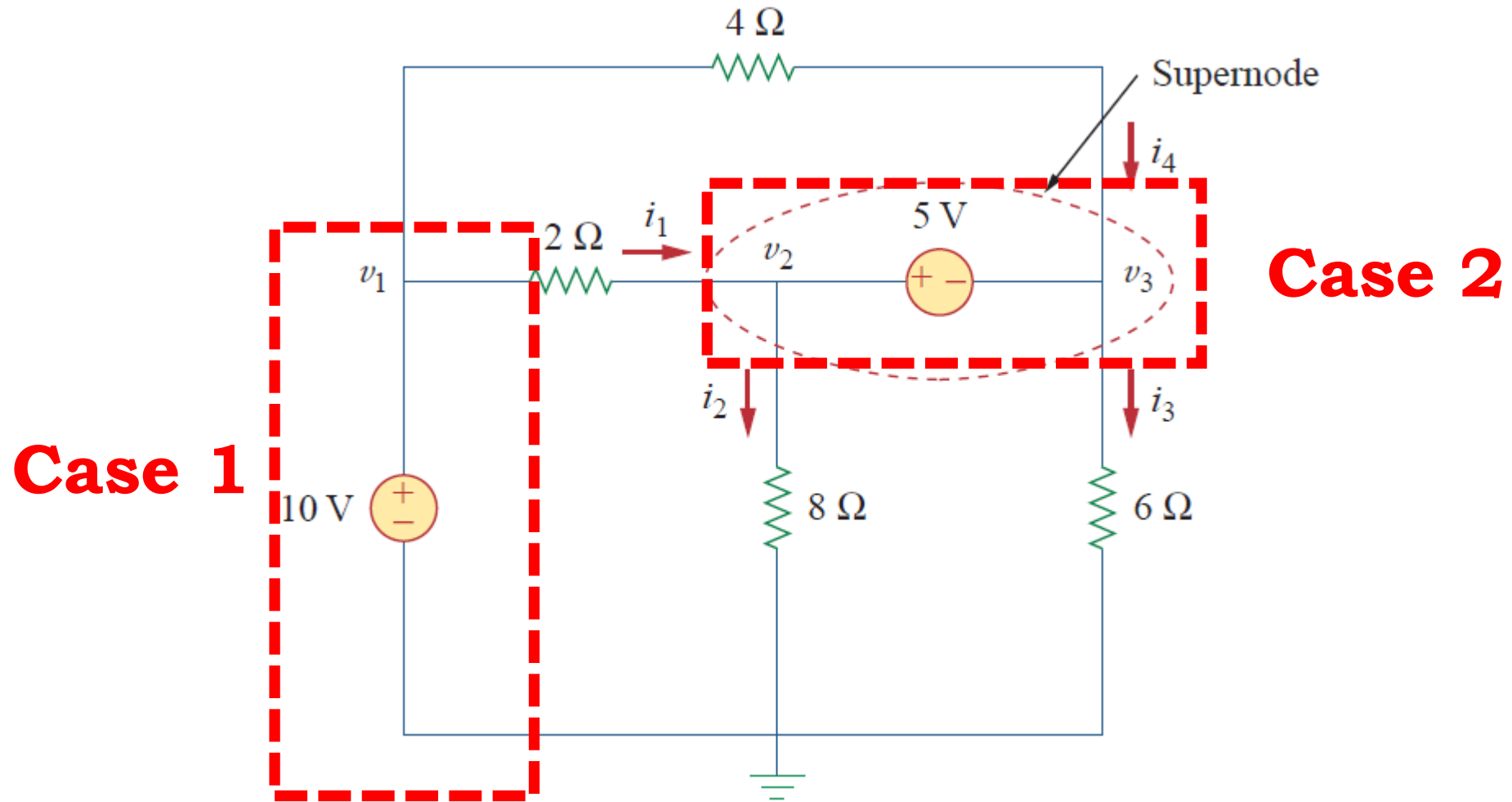


Nodal Analysis (Cont...)

- ▶ Calculate the node voltages in the circuit shown in figure below:



Nodal Analysis with Voltage Source



Nodal Analysis with Voltage Source (Cont...)

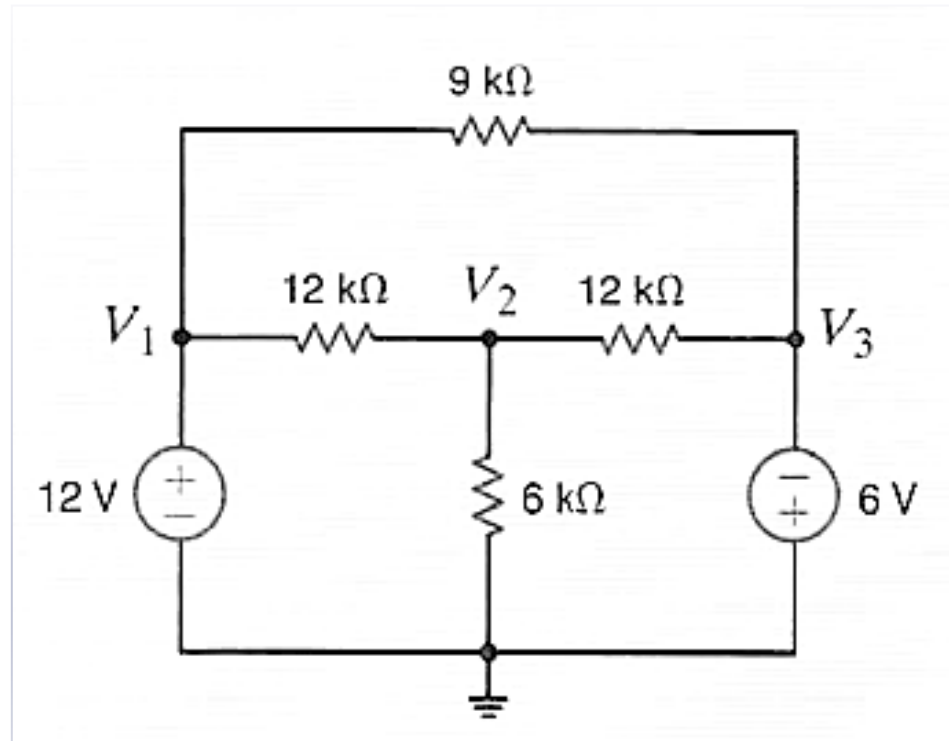
Case 1 :

If a voltage source is connected **between** the **reference node** and a **nonreference node**, we simply set the voltage at the nonreference node equal to the voltage of the voltage source.



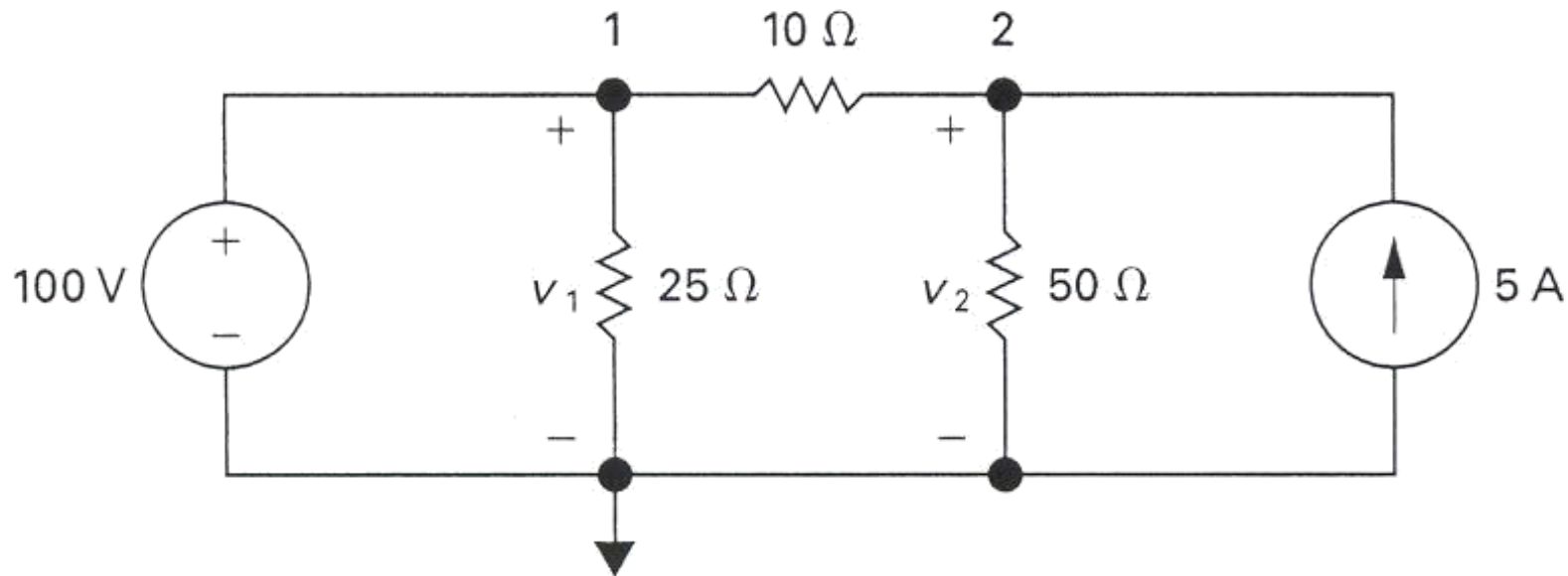
Nodal Analysis with Voltage Source (Cont...)

- Determine all node voltages:



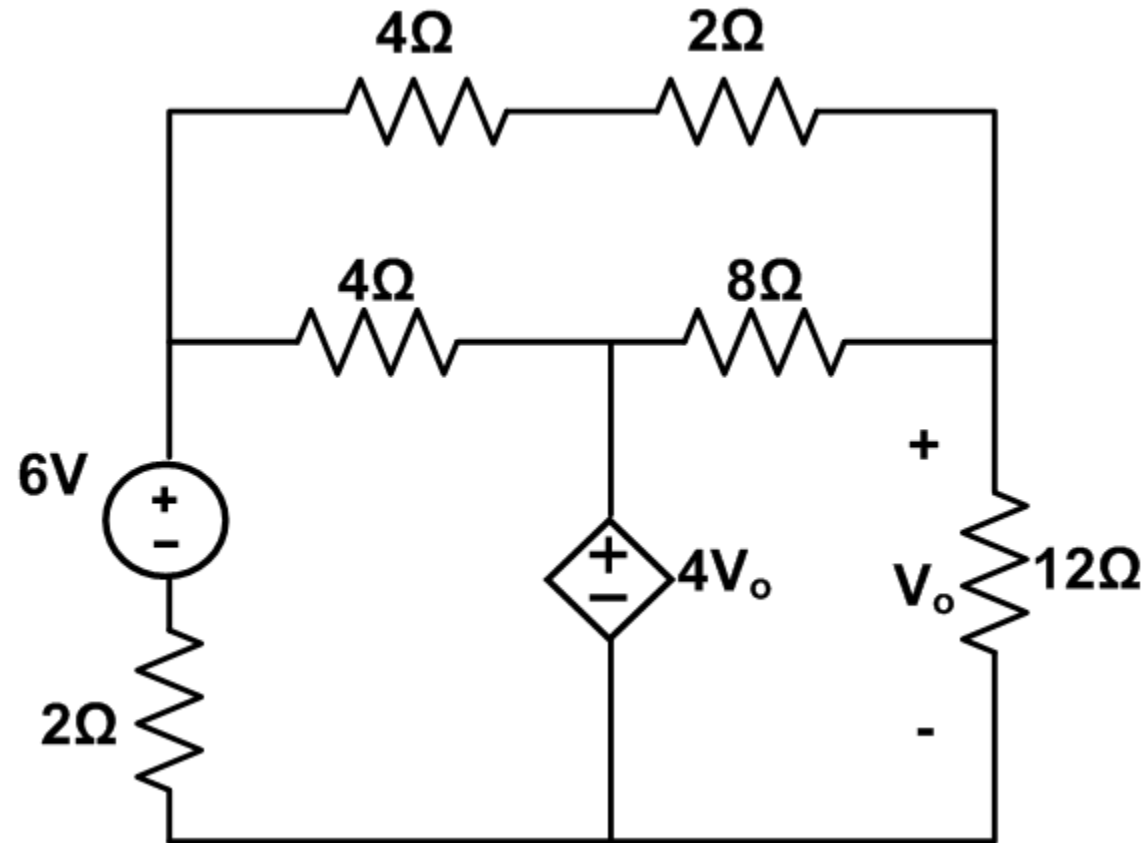
Nodal Analysis with Voltage Source (Cont...)

- Determine all node voltages:



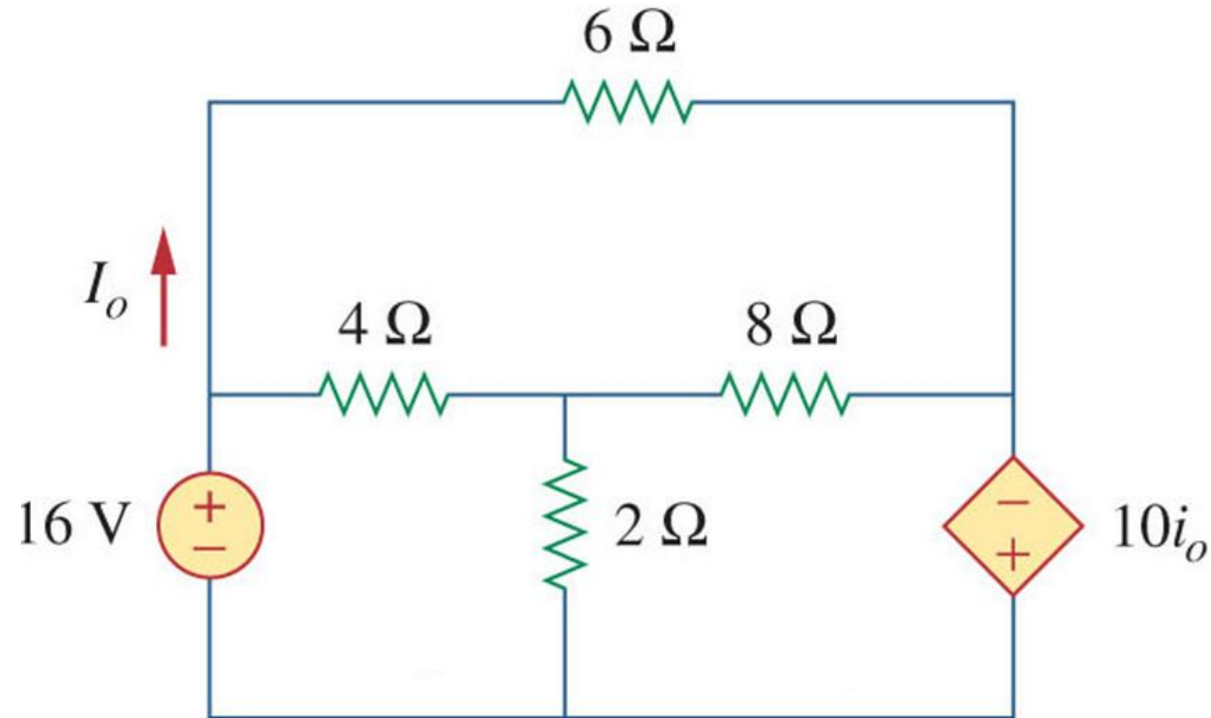
Nodal Analysis with Voltage Source (Cont...)

- Determine all node voltages:



Nodal Analysis with Voltage Source (Cont...)

- ▶ Using nodal analysis, find all node voltage and I_o :



Nodal Analysis with Voltage Source (Cont...)

Case 2 :

If the **voltage source** (dependent or independent) is connected between **two nonreference nodes**, the two nonreference nodes form a **generalized node or supernode**; we **apply** both **KCL and KVL** to determine the node voltages.

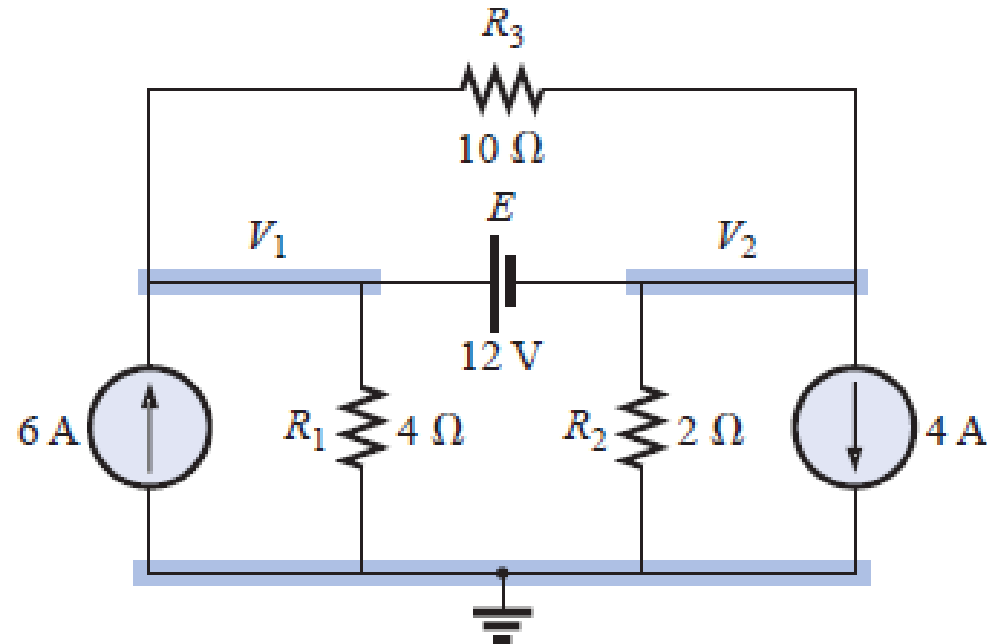
Supernode

- ▶ A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.



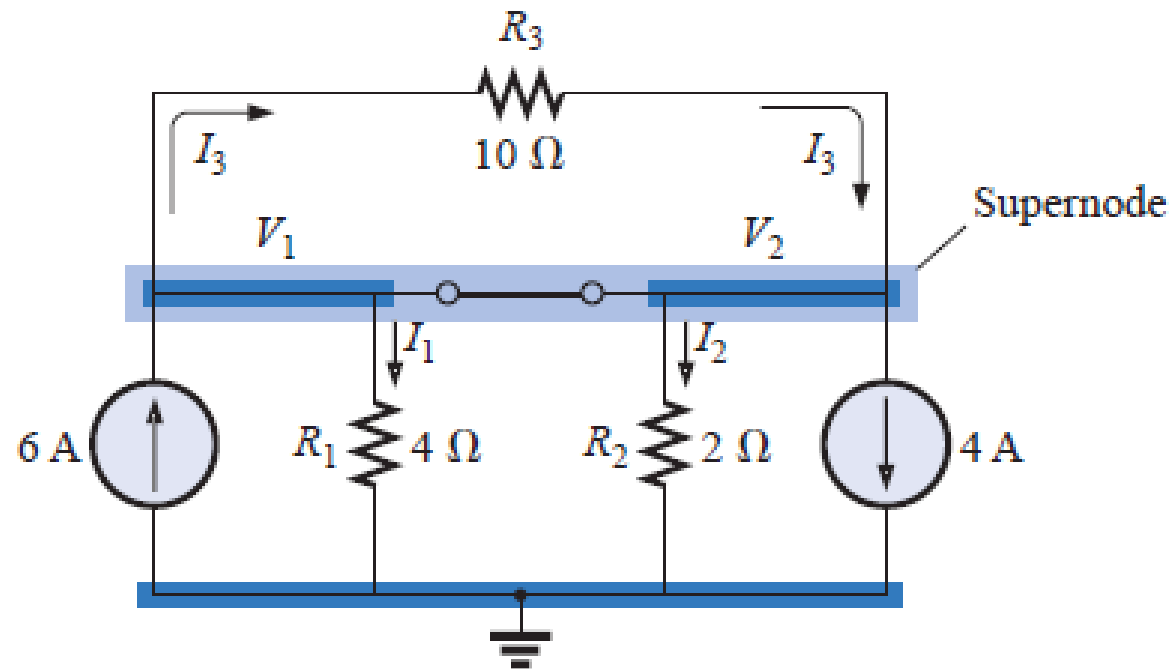
Supernode (Cont...)

- Determine the nodal voltages V_1 and V_2 of figure below using the concept of a supernode:



Supernode (Cont...)

- ▶ Replacing the independent voltage source of 12 V with a short-circuit equivalent



Supernode (Cont...)

Using KCL at supernode:

$$I_{6A} + I_3 = I_1 + I_2 + I_{4A} + I_3$$

$$I_{6A} - I_{4A} = I_1 + I_2$$

$$6 - 4 = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$2 = \frac{V_1}{4} + \frac{V_2}{2}$$

$$0.25V_1 + 0.5V_2 = 2$$

Voltage at supernode:

$$V_1 - V_2 = 12$$

Solving equation:

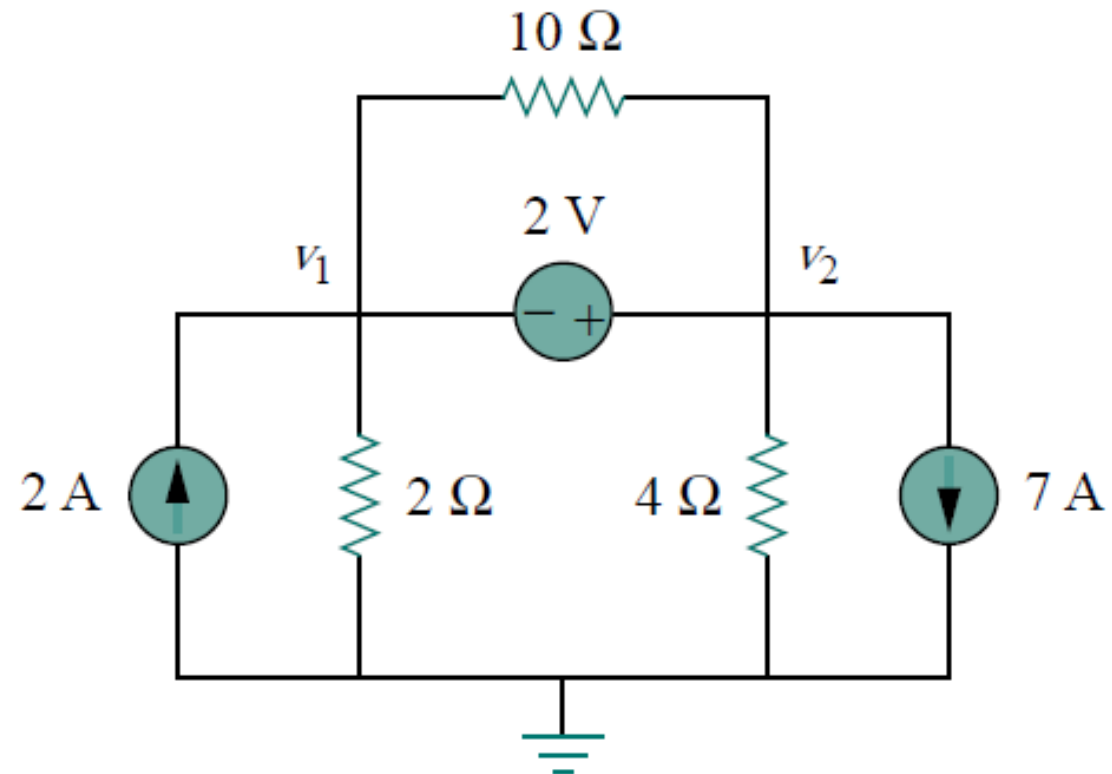
$$V_1 = 10.6667V$$

$$V_2 = -1.3333V$$



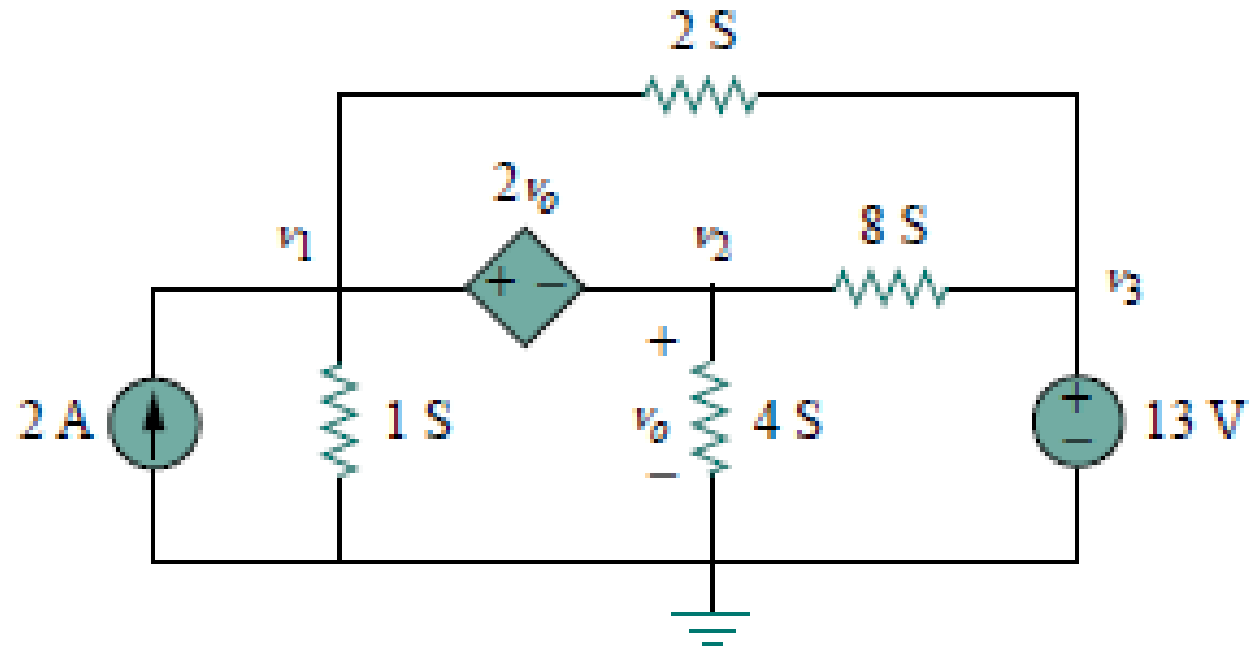
Supernode (Cont...)

- Find the node voltage:



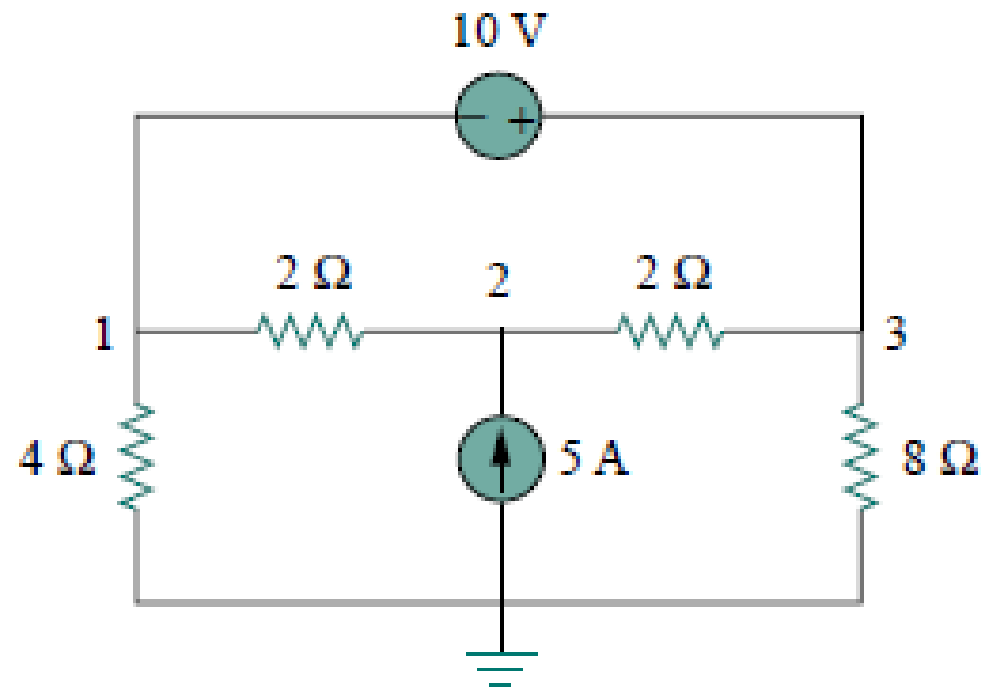
Supernode (Cont...)

- Find the node voltage:



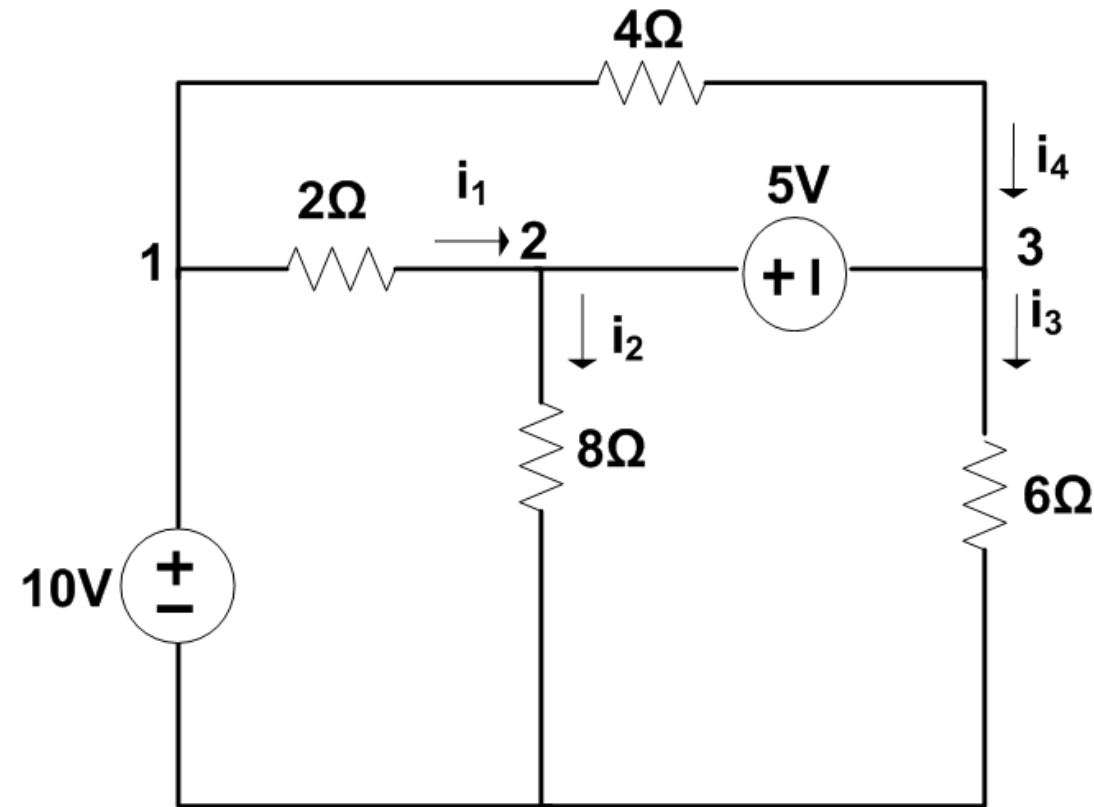
Supernode (Cont...)

- Find the node voltage:



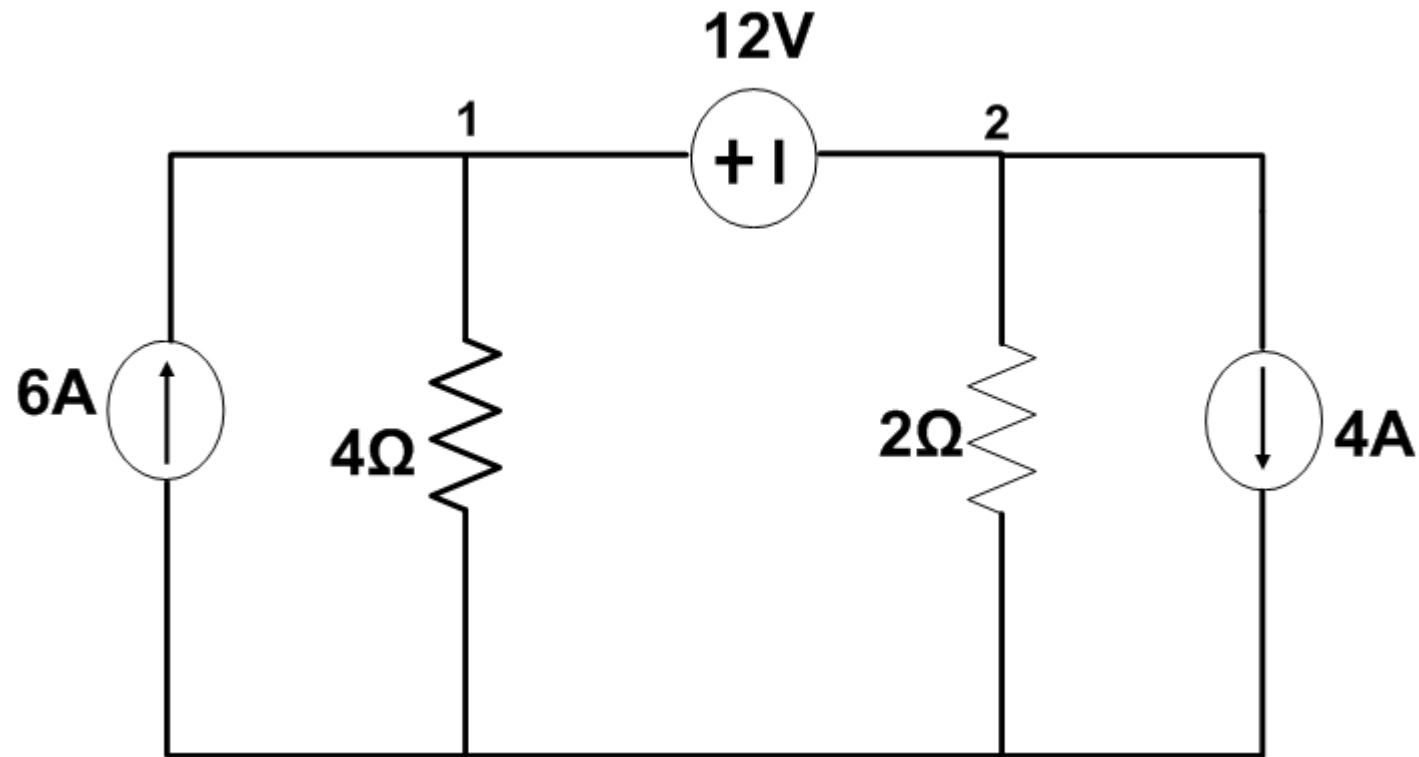
Supernode (Cont...)

- Find the node voltage:



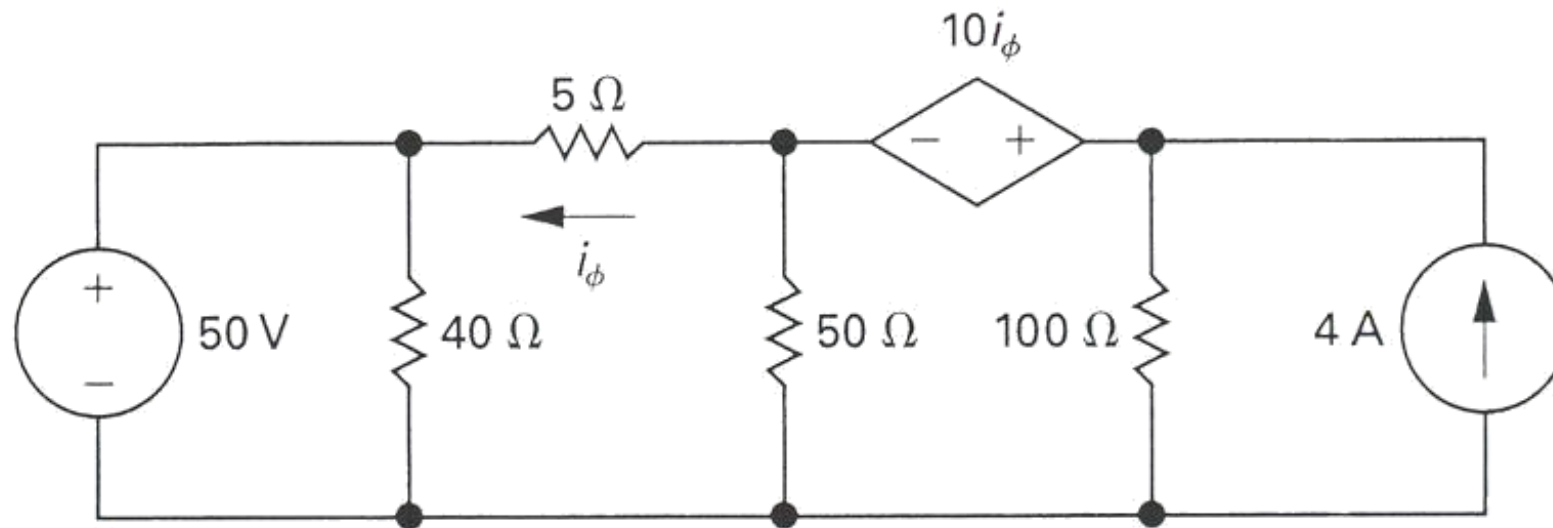
Supernode (Cont...)

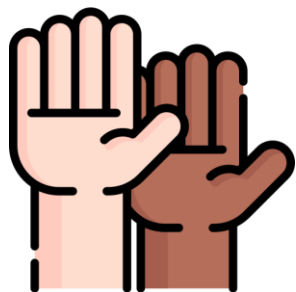
- Find the node voltage:



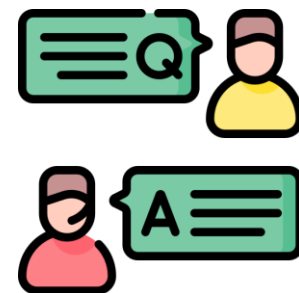
Supernode (Cont...)

- Find the node voltage:





Q & A



Contact



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